guidelines, and corporate goals. Its fundamental elements are:

Visibility: Acquiring thorough, up-to-date knowledge of cloud costs for all services, groups, and initiatives. Making educated decisions, recognising waste, and comprehending usage trends all depend on transparency.

Accountability: Giving particular groups or individuals charge of cost control, encouraging a proactive approach to cloud resource management, and cultivating a culture of ownership.

Optimisation: This is the process of continuously assessing and modifying resource use in order to cut down on wasteful spending, boost productivity, and optimise value.

Policy enforcement: Implementing rules and guardrails—such as budget limits, resource tagging, and access controls—to prevent overspending and ensure compliance with organisational standards.

An overview of the open source instruments for CCG

Because they allow for cost transparency without vendor lock-in and give businesses the freedom to customise solutions to meet their specific requirements, open source tools are essential for efficient cloud cost governance. They enable teams to modify and improve their cost management tactics by facilitating customisation and integration with current workflows.

Kubecost: A specialised open source tool called Kubecost was created to offer thorough cost allocation and monitoring in Kubernetes environments. Its main objective is to provide enterprises with real-time insight into the costs associated with their Kubernetes workloads, allowing teams to monitor expenditures at the namespace, deployment, or pod level. One of the main features is real-time cost allocation, which makes it easier to determine the precise location and cost of resource consumption. Additionally, it provides budget alerts, which enable proactive management by informing teams when expenses surpass predetermined thresholds.

OpenCost*:* OpenCost promotes transparency and uniformity across various platforms by acting as an open standard for cloud cost metrics. It offers a cloud-agnostic, community-driven API that standardises the gathering, reporting, and consumption of cost data. The relationship between OpenCost and Kubecost is fundamental; Kubecost ensures that cost data is interoperable and standardised by implementing the specifications established by OpenCost. OpenCost's primary benefit is its capacity to standardise cost metrics across various cloud environments and providers, dismantling silos and allowing businesses to handle expenses consistently.

Infracost: By offering infrastructure as code (IaC) cost estimates prior to deployment, Infracost closes a significant gap. During the development process, developers and DevOps teams can view a comprehensive cost breakdown of their infrastructure plans thanks to its seamless integration with well-known IaC tools like Terraform. Teams can avoid surprises during or after deployment by using this proactive insight to make cost-conscious decisions early. As part of the deployment process, Infracost automates cost checks through integration with CI/CD pipelines.

Architecture of workflow

Before any resources are provisioned, organisations use Infracost integrated with Terraform to estimate infrastructure costs as part of the pre-deployment (planning) stage of the cloud cost governance process. Teams can assess possible costs during the planning stage thanks to this proactive approach, which facilitates cost-conscious decision-making. Teams can automate cost checks as part of the deployment process by integrating Infracost into the CI/CD pipelines. This ensures that infrastructure plans are in line with financial constraints prior to deployment.

Workloads are deployed in Kubernetes clusters while Kubecost actively monitors the environment during the deployment (runtime monitoring) phase. Kubecost dynamically distributes costs according to actual resource consumption by retrieving real-time usage data from active pods, nodes, and namespaces. Teams can quickly spot possible inefficiencies or unanticipated cost increases thanks to this ongoing monitoring, which offers detailed insights into where and how money is being spent. By pointing out underutilised resources or downscaling opportunities, it also aids ongoing optimisation efforts.

The OpenCost API, which provides a standardised format for cost metrics across various environments and cloud providers, is utilised in the standardisation and reporting stage. Because of the interoperability and consistent reporting that this API guarantees, organisations can more easily centralise cost data, compare across various teams or projects, and produce thorough reports. Better tracking, analysis, and communication of cloud expenses are made possible by standardised metrics.

Step-by-step tool integration

A unified workflow for cloud cost management is produced by integrating Kubecost, OpenCost, and Infracost. Installing and setting up Kubecost on your Kubernetes cluster is the first step. This entails establishing data sources such as Prometheus, configuring access permissions, and deploying the Kubecost Helm chart or manifests. After installation, check Kubecost's functionality by going to its dashboard and making sure your workloads' real-time cost monitoring is displayed accurately.